1.5 SYMMETRY OPERATIONS

A *symmetry operation* is that which transforms the crystals to itself, i.e., a crystal remains invariant under a symmetry operation. These operations are *translation, rotation, reflection* and *inversion.* The translation operation applies to lattices only while all the remaining operations and their combinations apply to all objects and are collectively known as *point symmetry operations.* The inversion operation is applicable only to three-dimensional crystals. These operations are briefly described below:

(*i*) Translations

The translation symmetry follows from the orderly arrangement of a lattice. It means that a lattice point **r,** under lattice translation vector operation **T**, gives another point **r'** which is exactly identical to **r**, i.e.,

$$\mathbf{r'} = \mathbf{r} + \mathbf{T}$$

where **T** is defined by Eq. (1.1).

(*ii*) Rotations

A lattice is said to possess the rotation symmetry if its rotation by an angle θ about an axis (or a point in a two-dimensional lattice) transforms the lattice to itself. Also, since the lattice always remains invariant by a rotation of 2π, the angle 2π must be an integral multiple of θ, i.e.,

$$n\theta = 2\pi$$

or
$$\theta = 2\pi / n \qquad (1.3)$$

The factor n takes integral values and is known as *multiplicity of rotation axis.* The possible values of n which are compatible with the requirement of translation symmetry are 1, 2, 3, 4 and 6 only. Thus, for example, for n equal to 6, θ is 60° which means that the lattice repeats itself with a minimum rotation of 60°. Such a rotation is illustrated in Fig.1.5. Regular hexagon is an example of such a lattice. A rotation corresponding to the value of n is called n-fold rotation. A two-dimensional square lattice has 4-fold rotation symmetry. It may be noted that a rotation axis may or may not pass through a lattice point. The fact that 5-fold rotation is not compatible with translation symmetry operation and that only 1, 2, 3, 4 and 6 - fold rotations are permissible is proved as follows :

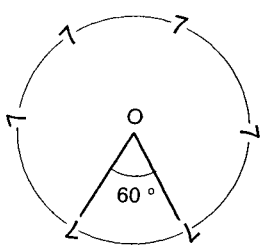


Fig. 1.5. *Six-fold rotation about the point O in two dimensions.*

Consider a row of lattice points A, B, C and D as shown in Fig. 1.6.

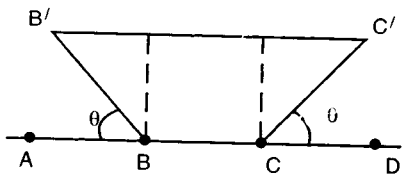


Fig. 1.6. *Geometry used to prove that only 1, 2, 3, 4 and 6 - fold rotation axes are permissible.*

Let **T** be the lattice translation vector and let the lattice have n - fold rotational symmetry with rotation axes passing through the lattice points perpendicular to the plane of paper. Rotations by an angle $\theta = 2\pi/n$ about points B and C in the clockwise and anticlockwise directions respectively yield points B' and C' which must be identical to B and C. Thus the points B' and C' must also be lattice points and should follow lattice translation symmetry. Hence B' C' must be some integral multiple of BC, i.e.,

$$B'C' = m(BC)$$

or
$$2T \cos\theta + T = mT$$

or
$$\cos\theta = (m - 1) / 2 \qquad (1.4)$$

where m is an integer. Since $|\cos\theta| \leq 1$, the allowed values of m are 3, 2, 1, 0 and –1. These correspond to the allowed values of θ as 0° or 360°, 60°, 90°, 120° and 180° respectively. Hence from Eq. (1.3), the permissible values of n are 1, 6, 4, 3 and 2. Thus we conclude that 5-fold rotation is not permissible because it is not compatible with lattice translation symmetry. Similarly, other rotations, such as 7-fold rotation, are also not permissible.

Figure 1.7 gives a convincing demonstration of non-existence of 5-fold rotation axis. As shown in the figure, the pentagons placed side by side do not cover the complete space. This is because no sets of vectors exist which satisfy translation symmetry operation throughout and hence this arrangement of pentagons cannot be regarded as a lattice. The array itself, however, has a 5-fold symmetry about the point A.

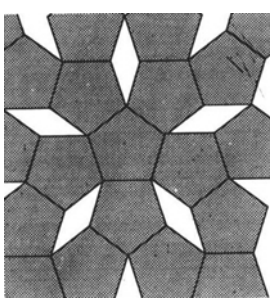


Fig. 1.7. *Demonstration of non-existence of a five-fold rotation axis in a lattice.*

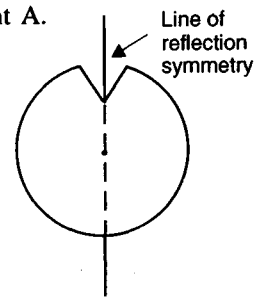


Fig. 1.8. *Reflection symmetry of a notched wheel about a line.*

(iii) Reflections

A lattice is said to possess reflection symmetry if there exists a plane (or a line in two dimensions) in the lattice which divides it into two identical halves which are mirror images of each other. Such a plane (or line) is represented by m. The reflection symmetry of a notched wheel is illustrated in Fig. 1.8. Considering the combinations of reflections with allowed rotations, we note that each allowed rotation axis can be associated with two possibilities : one is rotation with reflection and the other rotation without reflection. Since there are five allowed rotation axes, the possible number of such combinations is 10. These are designated as

$$1,\ 1m,\ 2,\ 2mm,\ 3,\ 3m,\ 4,\ 4mm,\ 6,\ 6mm$$

where the numerals represent the type of rotation axis, the first m represents a plane (or line) parallel to the rotation axis and the second m refers to another plane (or line) perpendicular to the rotation axis. These ten groups of symmetry operations are shown in Fig. 1.9.

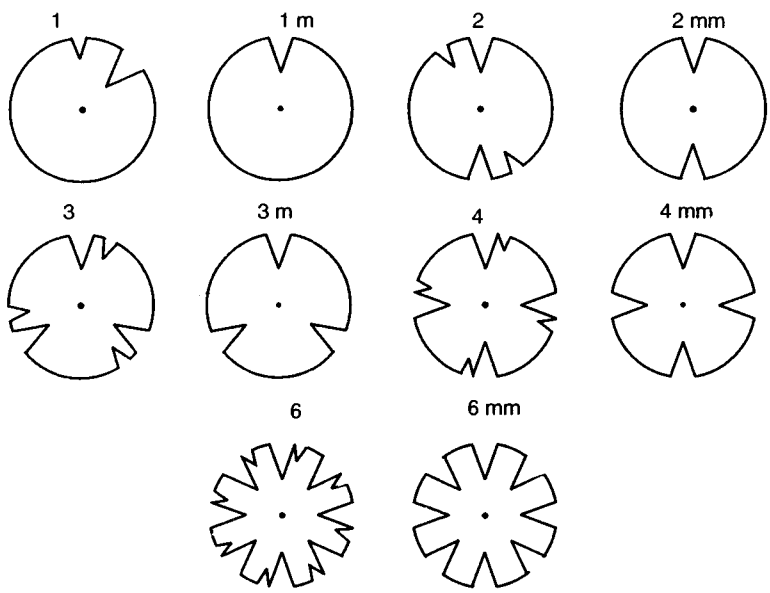


Fig. 1.9. *Ten two-dimensional point groups consisting of rotation and reflection symmetry operations illustrated using notched wheels.*

(iv) Inversions

Inversion is a point operation which is applicable to three-dimensional lattices only. This symmetry element implies that each point located at **r**

relative to a lattice point has an identical point located at $-\mathbf{r}$ relative to the same lattice point. In other words, it means that the lattice possesses a centre of inversion denoted by $\bar{1}$.

It may be noted that, apart from these symmetry operations, a three-dimensional lattice in particular may have additional symmetry operations formed by the combinations of the above-mentioned operations. One such example is *rotation-inversion operation.* These operations further increase the number of symmetry elements. These symmetry elements are further employed to determine the type of lattices possible in two and three-dimensional spaces.

1.6 POINT GROUPS AND SPACE GROUPS

We have seen that there are mainly four types of symmetry operations, i.e., translation, rotation, reflection and inversion. The last three operations are point operations and their combinations give certain symmetry elements which collectively determine the symmetry of space around a point. The group of such symmetry operations at a point is called a *point group.*

In two-dimensional space, rotation and reflection are the only point operations. As described earlier, their combinations yield 10 different point groups designated as 1, 1*m*, 2, 2*mm*, 3, 3*m*, 4, 4*mm*, 6, and 6*mm* which are shown in Fig. 1.9. In three-dimensional space, however, the situation is complicated due to the presence of additional point operations such as inversion. There are a total of 32 point groups in a three-dimensional lattice.

The crystals are classified on the basis of their symmetry which is compared with the symmetry of different point groups. Also, the lattices consistent with the point group operations are limited. Such lattices are known as *Bravais lattices.* These lattices may further be grouped into distinct crystal systems.

The point symmetry of crystal structure as a whole is determined by the point symmetry of the lattice as well as of the basis. Thus in order to determine the point symmetry of a crystal structure, it should be noted that

(*i*) a unit cell might show point symmetry at more than one locations inside it, and

(*ii*) the symmetry elements comprising combined point and translation operations might be existing at these locations.

The group of all the symmetry elements of a crystal structure is called *space group.* It determines the symmetry of a crystal structure as a whole. There are 17 and 230 distinct space groups possible in two and three dimensions respectively.